

National University of Computer and Emerging Sciences

FAST School of Computing

Fall-2022

Islamabad Campus

EE-2003 Computer Organization and Assembly Language (CS)

Serial No:

Sessional Exam-II**Total Time: 1 Hour****Total Marks: 60**Friday, 18th November, 2022.

Course Instructors

Dr. Niaz Ahmed, Ms. Sobia Rasheed, Ms. Shehr
Bano, Mr. Rohail Gulbaz, Mr. Shams Farooq

Signature of Invigilator

Student Name

Roll No.

Course Section

Student Signature

DO NOT OPEN THE QUESTION BOOK OR START UNTIL INSTRUCTED.

Instructions:

1. Attempt on question paper. Attempt all of them. Read the question carefully, understand the question, and then attempt it.
2. No additional sheet will be provided for rough work.
3. After being asked to commence the exam, please verify that you have TEN(10) different printed pages including this title page. There are a total of 6 questions.
4. Calculator sharing is strictly prohibited.
5. Use permanent ink pens only. Any part done using soft pencil will not be marked and cannot be claimed for rechecking.
6. Write your instructors name on top of the paper and share something good in the end of paper to score extra 3 marks.

	Q-1	Q-2	Q-3	Q-4	Q-5	Q-6	Total
Marks Obtained							
Total Marks	10	10	10	10	10	10	60

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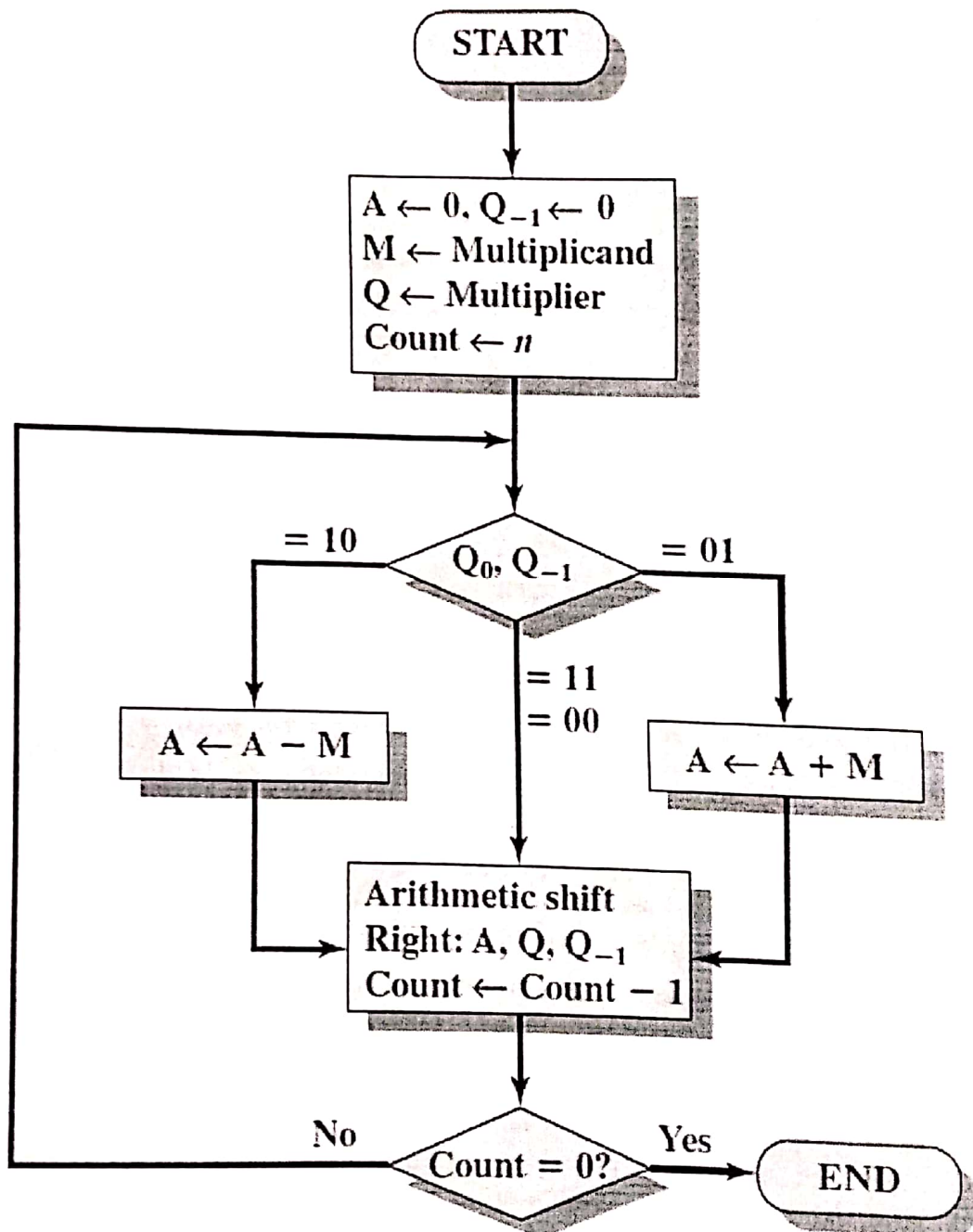
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Question 1 [10 Marks]

Use Booth's Multiplication Algorithm to multiply 23 (take it as Q) by -9 (take it as M). Show all steps considering your computer's data width equal to 6-bits.



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M (Multiplicand)

110111

$$\begin{array}{r} 001001 \\ 110111 \end{array} = -9$$

M

$$\begin{array}{r} 16 \quad 8 \quad 4 \quad 2 \quad 1 \\ 0 \quad 1 \quad 0 \quad 1 \quad 1 \end{array} = 23$$

A (Accumulator)	Q (Multiplier)	Q ₋₁	Operation
			Initial Values
000000	010111	0	
001001	010111	0	A - M (1)
000100	101011	1	Shift
000010	010101	1	Shift (2)
000001	001010	1	Shift (3)
111000	001010	1	A + M (4)
111100	000101	0	Shift
000101	000101	0	A - M (5)
000010	100010	1	Shift
111001	100010	1	A + M (6)
111100	110001	0	Shift
↓			
000011	001111		
128 64	84 21 = 207		

$$23 \times -9 = -207$$

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Question 2 [10 Marks]

Dry run the program given below and update the table for each iteration.

```
.model small
.data
    array db -5, 1, 0, 3
    swap db 0
.code
    mov ax, @data
    mov ds, ax

    mov cx, lengthof array
    dec cx

start:
    mov swap, 0
    mov bx, 0
loop1:
    mov dl, bl
    add dl, 48
    mov ah, 02
    int 21h

    mov al, [bx+array]
    cmp al, [bx+array+1]
    jbe noswap

    mov dl, [bx+array+1]
    mov [bx+array+1], al
    mov [bx+array], dl
    mov swap, 1

noswap:

    add bx, 1
    cmp bx, cx
    jne loop1

    cmp swap, 1
    je start

mov ah, 4ch
int 21h
end
```


Page 5 of 10

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Question 3 [10 Marks]

Carefully dry run the given program and write the output in Decimal. Also show complete traces of runtime stack for both fill and clear phases by writing the actual offset and values.

Offset	Code
	.model small
	.data 48 0 ↓ si
	n db <u>45</u> , 49, 52, 51, 53
	.stack 0100h
0000	.code
0001	mov ax, @data
0002	mov ds, ax
	jmp start
	MySub PROC uses dx cx
0003	
0004	push bp
0005	mov bp, sp
0006	
0007	sub sp, 4
	mov word ptr [bp-4], 59
0008	mov word PTR [bp-2], 57
0009	
000A	mov dl, [bp-2]
	mov ah, 02
000B	int 21h
000C	
000D	mov dx, [bp+4]
	mov ah, 02
000E	int 21h
000F	
0010	mov bx, [bp+10]
0011	mov dx, [bx]
	mov ah, 02
0012	int 21h
0013	mov sp, bp
	pop bp
0014	ret 2
	MySub ENDP
	start:
	main proc
0015	

00h 4B2
↑
LSB

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```

0016      mov si, offset n
0017      mov word ptr [si], 48
           add si, 2
0018
0019      mov dx, [si-2]
001A      mov ah, 02
           int 21h
001B
001C      mov dx, [si]
001D      mov ah, 02
           int 21h
001E
001F      mov di, offset n
0020      push di
0021      mov cx, 49
0022      push cx
           call MySub
0023
0024      inc cl
0025      push cx
0026      mov dx, cx
0027      mov ah, 02
           int 21h

           main endp

0028
0029      mov ah, 4ch
0030      int 21h
           end

```

dx = 52

Output:

049402

Fill Stack

Address	Content
0100	offset n = 0000h
00FE	49 50
00FC	ret add = 0029
00FA	dx = 52
00FB	cx = 49
00F6	bp = 0000
00F4	52
00F2	59
00F0	

← bp, sp

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Question 4 [10 Marks]

Update the values of flags after execution of each block which are part of a single program and write answers in the provided area. Rough Work is compulsory to update any flag.

<pre> .model small .code ;Block 1 mov al, 10101010b mov bl, 10101010b add bl, al ;Block 2 mov bl, 10101010b or al, bl ;Block 3 jne B cmp al, 1 B: cmp al, 0 </pre> <i>jne B True</i>	;Block 1 <table border="1"> <tr><td rowspan="6">Flags</td><td>Sign</td><td>0</td></tr> <tr><td>Zero</td><td>0</td></tr> <tr><td>Carry</td><td>1</td></tr> <tr><td>Overflow</td><td>1</td></tr> <tr><td>Parity</td><td>0</td></tr> <tr><td>Auxiliary</td><td>1</td></tr> </table> ;Block 2 <table border="1"> <tr><td rowspan="6">Flags</td><td>Sign</td><td>1</td></tr> <tr><td>Zero</td><td>0</td></tr> <tr><td>Carry</td><td>0</td></tr> <tr><td>Overflow</td><td>0</td></tr> <tr><td>Parity</td><td>1</td></tr> <tr><td>Auxiliary</td><td>1</td></tr> </table> ;Block 3 <table border="1"> <tr><td rowspan="6">Flags</td><td>Sign</td><td>1</td></tr> <tr><td>Zero</td><td>0</td></tr> <tr><td>Carry</td><td>0</td></tr> <tr><td>Overflow</td><td>0</td></tr> <tr><td>Parity</td><td>1</td></tr> <tr><td>Auxiliary</td><td>0</td></tr> </table>	Flags	Sign	0	Zero	0	Carry	1	Overflow	1	Parity	0	Auxiliary	1	Flags	Sign	1	Zero	0	Carry	0	Overflow	0	Parity	1	Auxiliary	1	Flags	Sign	1	Zero	0	Carry	0	Overflow	0	Parity	1	Auxiliary	0	Rough Work (Compulsory) 1 10101010 + 10101010 ----- 01010100 CF=1 AF=1 10101010 10101010 ----- 10101010 10101010 - 00000000 ----- 10101010
Flags	Sign		0																																						
	Zero		0																																						
	Carry		1																																						
	Overflow		1																																						
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	Auxiliary	0																																							

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Question 5 [10 Marks]

Find the value of AL in Decimal for the following set of instructions.

mov ax, -70 shr ax, 1	AL= -35	mov ax, -70 shr al, 1	AL= 93
mov ax, 0 mov al, -70 shr ax, 1	AL= 93	mov al, -70 shr al, 1	AL= 93
mov ax, -70 sar ax, 1	AL= -35	mov ax, -70 sar al, 1	AL= -35
mov ax, 0 mov al, -70 sar ax, 1	AL= 93	mov al, -70 sar al, 1	AL= -35

Rough Work:

$$\begin{array}{r}
 2 \overline{) 70} \\
 \underline{2 35} \\
 2 17 \\
 \underline{2 8} \\
 2 4 \\
 \underline{2 2} \\
 2 1
 \end{array}$$

$$- \dots 01000110$$

$$1111111 \cdot 10111010 = -70$$

$$\begin{array}{r}
 11011101 = -35 \\
 00100011 \\
 32 \quad 21
 \end{array}$$

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Question 6 [7 +3 Marks]

SHL instruction performs unsigned multiplication when the multiplier is a power of 2 and any other number can be expressed in powers of 2. Write instructions to find the product of AL by 29, where AL=4. A sample is provided below.

Sample Program: AX x 36

```
mov ax, 123
mov bx, ax
shl ax, 5
shl bx, 2
add ax, bx
```

Problem: AL=4, AL * (29)

AL can have any value

```
mov al, 4
mov bl, al
mov cl, al
mov dl, al

shl bl, 4
shl cl, 3
shl dl, 2

add al, bl
add al, cl
add al, dl
```

AL x 4
29 x 4 is wrong

16	8	4	2	1	
1	1	1	0	1	0
4	3	2	2	2	2
2	2	2	2	2	2

Good Luck....

3 marks of writing anything good more specifically about your teacher. and mentioning the name of your teacher.